

FANR 3800: Weekly ArcGIS Pro Tools & Methods

Tools and methods introduced each week (ArcGIS Pro v3.7x)

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1 Weekly ArcGIS Pro Tools & Methods

Tracks the ArcGIS Pro **tools** and the **methods/concepts** *first introduced* each week (not cumulative). Target software: **ArcGIS Pro v3.7x**.

1.1 Notation Convention (Applies to All Course Documents)

- **ArcGIS Pro tools** are written in ALL-CAPS INSIDE BACKTICKS, e.g. BUFFER, SELECT BY LOCATION, RASTER CALCULATOR.
- **Methods and concepts** (things you *do* or *understand*, not a specific tool) are written in *Proper-Case Italics*, e.g. *Digitizing*, *Attribute Query*, *Map Algebra*.

1.2 Verification Note

Tool names below are standard ArcGIS Pro geoprocessing tool names. **Verify spelling, toolbox location, license level, and the PAIRWISE vs. classic variant against your v3.7x install** before finalizing, these change across versions. Items marked *(ribbon/pane)* are interactive workflows on a ribbon or pane rather than a single geoprocessing tool.

1.3 Arc A: Build a Base You Can Trust

Wk	ArcGIS Pro tools (ALL-CAPS)	Methods & concepts (<i>Proper-Case italic</i>)
1	<i>(none, project creation & interface navigation)</i>	<i>Spatial Thinking, Vector Data Model, Raster Data Model</i>
2	<i>(none, SYMBOLOGY pane & LAYOUT view are ribbon/pane; EXPORT LAYOUT)</i>	<i>Symbology (Categorized, Graduated, Unique Values), Map Layout, Cartographic Communication</i>
3	PROJECT, DEFINE PROJECTION	<i>Coordinate System Specification, Datum Transformation, Projection Selection</i>
4	CREATE FEATURES <i>(editing, ribbon/pane)</i> , GPX TO FEATURES, XY TABLE TO POINT, GEOREFERENCE <i>(ribbon toolset)</i>	<i>Digitizing, Vector Editing, GPS Data Collection (Static, Dynamic), Air-Photo Interpretation, Georeferencing</i>

1.4 Arc B: Ask Questions of Your Data

Wk	ArcGIS Pro tools (ALL-CAPS)	Methods & concepts (<i>Proper-Case italic</i>)
5	SELECT BY ATTRIBUTES (SELECT LAYER BY ATTRIBUTE), SELECT BY LOCATION (SELECT LAYER BY LOCATION), ADD JOIN, CALCULATE GEOMETRY, SUMMARY STATISTICS	<i>Attribute Query, Spatial Query, Table Join, Inventory Summary, Area Calculation</i>

1.5 Arc C: Where Should Something Happen? (Vector)

Wk	ArcGIS Pro tools (ALL-CAPS)	Methods & concepts (<i>Proper-Case italic</i>)
6	BUFFER (PAIRWISE BUFFER), CLIP (PAIRWISE CLIP), ERASE (PAIRWISE ERASE)	<i>Proximity Analysis, Constraint Application, Feature Extraction, Dissolve</i>
7	INTERSECT (PAIRWISE INTERSECT), UNION, UPDATE	<i>Topological Overlay, Cross-Tabulation, Multi-Criteria Combination</i>
8	MODELBUILDER (<i>workflow documentation; optional</i>)	<i>Workflow Decomposition, Problem-to-Workflow Translation, Analysis Verification, Lecture Exam 1 (Wed)</i>

1.6 Arc D: Conditions That Vary Across the Land (Raster)

Wk	ArcGIS Pro tools (ALL-CAPS)	Methods & concepts (<i>Proper-Case italic</i>)
9	(<i>none, raster display & RASTER PROPERTIES; Spatial Analyst environment</i>)	<i>Raster Data Model (in depth), Spatial Resolution, Cell Size Selection, Lab Exam 1 (Wed/Thu)</i>
10	RECLASSIFY, EUCLIDEAN DISTANCE	<i>Reclassification, Distance Surface, Binary Suitability</i>
11	SLOPE, ASPECT, HILLSHADE	<i>Terrain Analysis, Surface Derivatives</i>
12	RASTER CALCULATOR, WEIGHTED OVERLAY	<i>Map Algebra, Weighted Overlay, Suitability Modeling</i>
13	CLASSIFICATION WIZARD (<i>Imagery ribbon</i>), CLASSIFY RASTER, TRAIN ... CLASSIFIER	<i>Image Interpretation, Supervised Classification, Unsupervised Classification, Change Detection</i>

1.7 Arc E: Real Problems Mix Everything

Wk	ArcGIS Pro tools (ALL-CAPS)	Methods & concepts (<i>Proper-Case italic</i>)
14	FEATURE TO RASTER, POLYGON TO RASTER, RASTER TO POLYGON, EXTRACT BY MASK	<i>Data Model Conversion, Integrated Workflow, Scenario Analysis</i> , Lecture Exam 2 (Wed); Lab Exam 2 (Wed/Thu)
15	<i>(Thanksgiving, no class)</i>	-
16	<i>(none, synthesis & course overview)</i>	<i>Method Selection & Sequencing, Synthesis</i>

1.8 Quick Alphabetical Tool Index

For fast lookup of where each tool first appears.

Tool	First introduced
ADD JOIN	Wk 5
ASPECT	Wk 11
BUFFER / PAIRWISE BUFFER	Wk 6
CALCULATE GEOMETRY	Wk 5
CLASSIFICATION WIZARD / CLASSIFY RASTER	Wk 13
CLIP / PAIRWISE CLIP	Wk 6
CREATE FEATURES	Wk 4
DEFINE PROJECTION	Wk 3
ERASE / PAIRWISE ERASE	Wk 6
EUCLIDEAN DISTANCE	Wk 10
EXTRACT BY MASK	Wk 14
FEATURE TO RASTER	Wk 14
GEOREFERENCE	Wk 4
GPX TO FEATURES	Wk 4
HILLSHADE	Wk 11
INTERSECT / PAIRWISE INTERSECT	Wk 7
MODELBUILDER	Wk 8
POLYGON TO RASTER	Wk 14
PROJECT	Wk 3
RASTER CALCULATOR	Wk 12
RASTER TO POLYGON	Wk 14
RECLASSIFY	Wk 10
SELECT BY ATTRIBUTES	Wk 5
SELECT BY LOCATION	Wk 5
SLOPE	Wk 11

Tool	First introduced
SUMMARY STATISTICS	Wk 5
UNION	Wk 7
UPDATE	Wk 7
WEIGHTED OVERLAY	Wk 12
XY TABLE TO POINT	Wk 4